IBM SkillsBuild - AICTE BharatCares Machine Learning & Applied AI Internship 2026

LEAN CANVAS BUSINESS & OPERATIONAL MODEL

Project Name: NeuroVision AI
Focus Domain: Healthcare & Medical Artificial Intelligence (SDG 3 & SDG 9)
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Cohort: Machine Learning & Applied AI (August - September 2026)

📊 Visual Lean Canvas (9-Box Industry Matrix)

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| 1. PROBLEM                       | 4. SOLUTION                      | 3.
UNIQUE VALUE PROPOSITION       | 9. UNFAIR ADVANTAGE              | 2. CUSTOMER
SEGMENTS              |
|                                  |                                  |
|                                  |                                  |
| • Critical shortage of certified  | • 4-class transfer learning model | "24/7
Expert Neuroradiology       | • Proprietary two-stage fine-     | • Primary:
Diagnostic             |
|   neuroradiologists (1:100,000 in |   (EfficientNetB0) classifying    |
Second-Opinion with Visual       |   tuned EfficientNetB0 pipeline   |
Radiologists & Neuroradiologists|
|   rural/semi-urban regions).      |   Glioma, Meningioma, Pituitary,  |  Grad-
CAM Confirmation and        |   incorporating CLAHE medical     | • Secondary:
Neurosurgeons &      |
| • Diagnostic fatigue causing      |   and Normal brain parenchyma.    |
Automated PDF Medical Reports    |   contrast enhancement.           |   Neuro-
Oncologists               |
|   10–25% interpretation error     | • Real-time Grad-CAM attention    |  in
under 150 Milliseconds."      | • Real-time Grad-CAM explainable  | • Tertiary:
Regional Diagnostic   |
|   rates in complex MRI studies.   |   heatmaps delineating tumor.     |
|   AI tightly coupled with vector  |   Imaging Centers & Tele-radiology|
| • "Black-box" AI skepticism due   | • Automated hospital-grade PDF    | High-
Level Concept:               |   ReportLab PDF generation.       |   Networks
|
|   to lack of visual explanations. |   radiology diagnostic reports.   | An
intelligent, transparent       | • Validated on 7,200 curated      |
|
|                                  |                                  |
clinical co-pilot reducing MRI    |   clinical MRI scans with         | Early
Adopters:                   |
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Existing Alternatives:		review latency by 60%.	>96% test accuracy.	Private diagnostic chains &
Manual reading; slow outsourced tele-radiology; unexplainable CAD.	emergency trauma triage centers.	

8. KEY METRICS 5. CHANNELS	
• Model Accuracy (>96%) & AUC (>0.98) across 4 classes. • Direct PACS / RIS integration via RESTful API endpoints. • Diagnostic Latency (<150 ms/scan • Tele-radiology enterprise cloud platforms (Render, AWS). • Radiologist triage time savings (target: 60% reduction). • Medical technology expos & radiological conferences (RSNA). • Daily active scans processed & patient report downloads. • IBM SkillsBuild / AICTE showcase	

7. COST STRUCTURE	6. REVENUE STREAMS
• Cloud GPU/CPU Compute & Storage (AWS / Render / DigitalOcean) • Data pipeline maintenance, dataset acquisition & model retraining • Medical regulatory audit, HIPAA/GDPR data compliance validation • Full-stack engineering, continuous integration & clinical support	• Per-Scan SaaS Subscription (e.g., ₹50 / $0.60 per diagnostic scan) • Annual Enterprise Hospital License (unlimited on-premise Docker) • B2B Tele-radiology API Platform Licensing • Academic & Research Institution Training Partnerships

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📝 Detailed Block-by-Block Analysis

1. Problem

- **Neuroradiologist Bottleneck:** Rural hospitals and district diagnostic centers often wait 24 to 72 hours for an outsourced radiologist to interpret brain MRI studies.
- **Diagnostic Inconsistency:** Micro-adenomas or subtle meningiomas in early stages are easily missed during high-volume overnight emergency shifts.
- **Explainability Gap:** Clinicians reject traditional machine learning models because they cannot verify *why* a neural network arrived at a particular classification.

2. Customer Segments

- **Hospitals & Imaging Centers:** High-volume facilities seeking automated triaging to prioritize urgent high-grade gliomas ahead of routine scans.
- **Tele-Radiology Networks:** Virtual diagnostics platforms looking for automated pre-reads and structured draft reporting.
- **Clinicians in Remote Centers:** Primary care practitioners who need rapid confirmation before airlifting or transferring acute neurosurgical patients.

3. Unique Value Proposition (UVP)

- **Instant Medical Validation:** In under 150 milliseconds, clinicians receive a 4-class classification, a Grad-CAM visual attention heatmap overlaid on the MRI, and a board-certified PDF diagnostic report ready for immediate printing or sharing.

4. Solution

- **EfficientNetB0 Backbone:** Achieves exceptional transfer learning performance with only 5.3 million parameters.
- **Grad-CAM Integration:** Uses gradient backpropagation to the `top_conv` layer to visually pinpoint the exact spatial boundaries driving the diagnosis.
- **Full Clinical Stack:** Complete with patient history management, search filtering, Chart.js analytics, and automated ReportLab PDF reports.

5. Channels

- **Cloud & On-Premise Deployment:** Docker containers ready for local hospital intranets or secure healthcare cloud services.
- **REST API Endpoints:** Allows direct programmatic ingestion from DICOM PACS servers (`POST /api/predict`).
- **Academic & Healthcare Partnerships:** Showcased through AICTE and IBM SkillsBuild healthcare innovation tracks.

6. Revenue Streams

- **Volume-Based SaaS:** Affordable pay-per-scan model for outpatient clinics.
- **Enterprise Licensing:** Fixed annual fee for multi-specialty hospital networks with on-premise data residency guarantees.

- **White-Label API:** Integration into existing electronic health record (EHR) and tele-radiology systems.

7. Cost Structure

- **Compute Infrastructure:** Cloud GPU/CPU inference servers.
- **Compliance & Security:** Ensuring HIPAA, GDPR, and DICOM data confidentiality.
- **Engineering & Support:** Continuous model monitoring, retraining with new clinical cohorts, and platform updates.

8. Key Metrics

- **Clinical Performance:** Test Accuracy $\geq 96.0\%$, Macro F1 ≥ 0.96, Macro AUC ≥ 0.98.
- **Speed:** Total response time under 150 milliseconds per scan.
- **Productivity:** 60% reduction in radiologist initial triage and documentation drafting time.

9. Unfair Advantage

- **End-to-End Synergy:** Seamless integration of CLAHE preprocessing, compound scaled EfficientNetB0, Grad-CAM feature attribution, SQLite persistence, and vector PDF medical reporting in a single unified, production-ready codebase.